CS 486/686
Decision Trees

Yuntian Deng

Lecture 16

RN 19.3, 19.8 · PM 7.3.1, 7.5

Reminder: next week is asynchronous

> 🎥 I'm away at **ICML** next week, so **L17 (Tue Jul 7)** and **L18 (Thu Jul 9)** are **pre-recorded videos** — **no in-person class** those days. Watch on your own time (links posted before each class).
>
> 📅 **Deadlines are unchanged:** Chat 8 + the CS686 project proposal are due **Tue Jul 7**; Chat 9 is out and **Assignment 2 is due Thu Jul 9**.
>
> 💬 Questions? Post on **Piazza** — the TAs are available all week, and I'll follow up when I'm back.

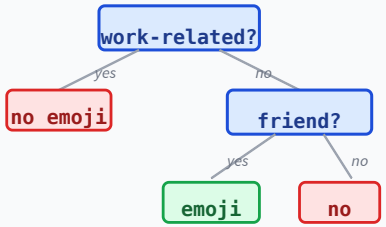
Learning goals

- Describe the **components** of a decision tree.
- **Construct** a tree from a feature-testing order.
- Compute a tree's **test-set accuracy**.
- Compute **entropy** and **information gain**; trace the **DT learner**.
- Use **Gini** / **CART**; control overfitting with **pruning**.
- Combine trees: **random forests** and **gradient boosting**.

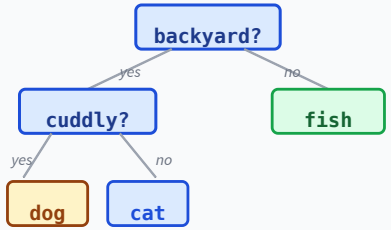
Decision trees in the wild

A decision tree is nothing more than a **nested if-then-else** — learned automatically from data.

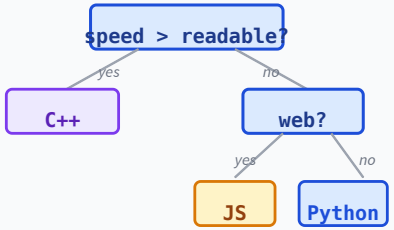
Should I use an emoji?



What pet?



Which language?



Simple to read, fast to evaluate, and surprisingly effective in practice.

Running example: Jeeves the valet

Predict from the weather: will Bertie play tennis today? Jeeves has logged 14 mornings.

Features and target

- **Features:** Outlook (Sunny/Overcast/Rain), Temp (Hot/Mild/Cool), Humidity (High/Normal), Wind (Weak/Strong).
- **Target:** Tennis? (Yes / No).

Learn from the training days → evaluate on a held-out test set.

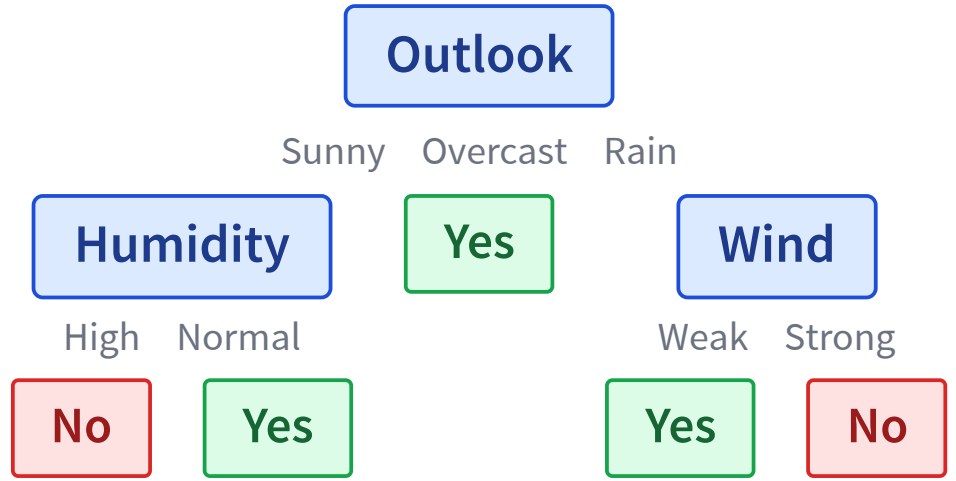
The training set (14 days)

Day	Outlook	Temp	Humidity	Wind	Tennis?
1	Sunny	Hot	High	Weak	No
2	Sunny	Hot	High	Strong	No
3	Overcast	Hot	High	Weak	Yes
4	Rain	Mild	High	Weak	Yes
5	Rain	Cool	Normal	Weak	Yes
6	Rain	Cool	Normal	Strong	No
7	Overcast	Cool	Normal	Strong	Yes
8	Sunny	Mild	High	Weak	No
9	Sunny	Cool	Normal	Weak	Yes
10	Rain	Mild	Normal	Weak	Yes
11	Sunny	Mild	Normal	Strong	Yes
12	Overcast	Mild	High	Strong	Yes
13	Overcast	Hot	Normal	Weak	Yes
14	Rain	Mild	High	Strong	No

9 Yes, 5 No. We'll learn a tree from these 14 rows.

Anatomy of a decision tree

- A supervised classifier with a **single discrete target**.
- Each **internal node** tests an input feature.
- **Edges** are labeled with feature values.
- Each **leaf** predicts a target class.
- **Classify**: follow edges from the root down to a leaf.



Sketch only — we'll build the real tree shortly.

Classify an example: walk down the tree

Given a new example, follow the tests from the root to a leaf:

> Example: **Outlook = Overcast, Temp = Hot, Humidity = High, Wind = Weak.**
>
> Test Outlook $\longrightarrow$ value is Overcast $\longrightarrow$ leaf **Yes**. Done.

Observation: the path itself is a nested if-then-else program. Converting a tree to code is straightforward.

Build a tree given an order

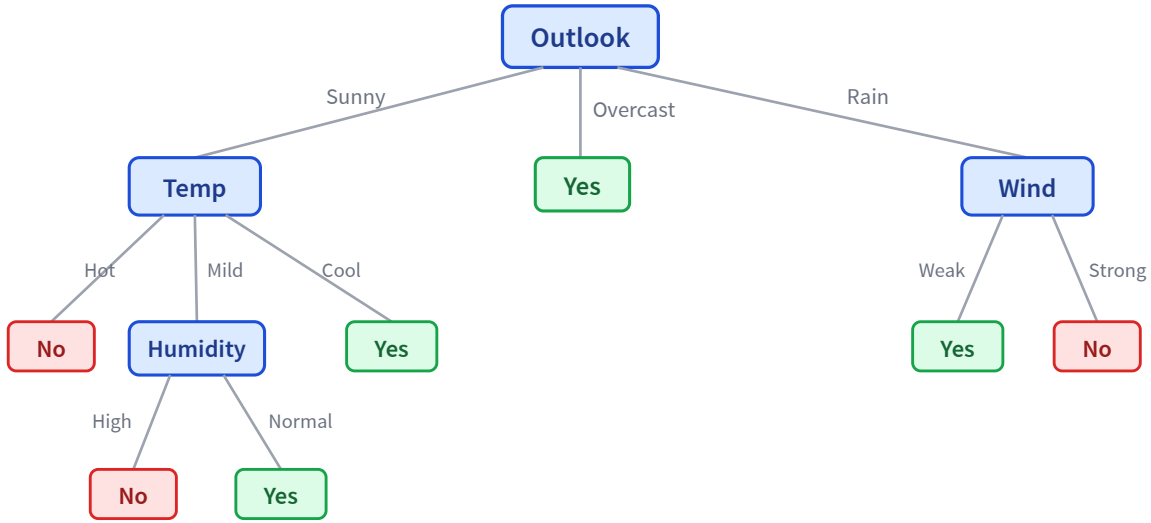
Pick a feature-testing order; recursively split the training examples by that feature's value.

An order for the Jeeves data

- **Root:** test Outlook.
- Sunny → test **Temp** (then Humidity if still mixed).
- Overcast → all Yes, done.
- Rain → test **Wind**.

Each example flows down the tree according to its feature values; leaves end up with subsets of the training set.

The Jeeves decision tree



Built from the 14 training rows by following the feature order above.

When do we stop? Three cases

All same class

Every example at this node has the same label.

Return that label.

No features left

We've tested every feature, but examples still disagree (noisy data).

Return the **majority class** at this node.

No examples left

A feature value unseen in training — no examples reach here.

Return the **parent's majority class**.

"No features left" → noisy labels; "no examples left" → unseen feature combinations.

The decision-tree learner

`DT-Learner(examples, features)`

1. If all examples are in the same class, return that class.
2. Else if no features left, return the majority class of *examples*.
3. Else if no examples left, return the majority class at the *parent*.
4. Else: choose a feature f; for each value v of f,
   - build an edge labeled v;
   - recurse on examples with $f = v$ and features $\setminus \{f\}$.

The only open question: **which feature do we pick to test?**

Which feature should we test?

Pick the feature whose split makes the children most **class-pure**.

Split on Outlook

Overcast cleanly predicts **Yes** — one branch is pure!

Split on Temp

Every Temp value still mixes Yes/No — messier.

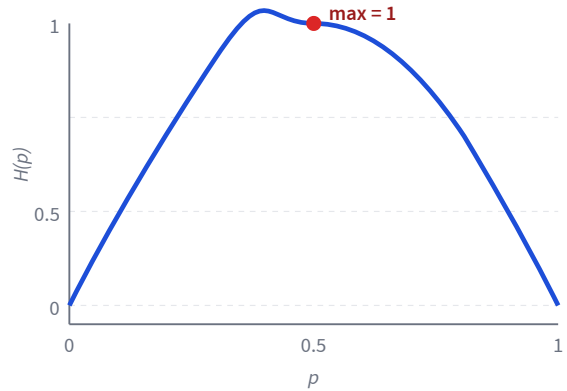
We need a metric to score "purity". Enter **entropy**.

Entropy: measuring uncertainty

Entropy

$$I\big(P(c_1), \ldots, P(c_k)\big) = -\sum_{i=1}^{k} P(c_i)\, \log_2 P(c_i)$$

Bits of uncertainty. Bigger = less certain.



- $I(0.5, 0.5) = 1$: maximally uncertain.
- $I(0.01, 0.99) \approx 0.08$: nearly certain the second outcome.
- $I(1, 0) = I(0, 1) = 0$: no uncertainty at all.

Expected information gain

Split on a feature with k values → compare the entropy *before* vs. the weighted entropy *after*:

before (parent): $H_{\text{before}} = I\left(\frac{p}{p+n}, \frac{n}{p+n}\right)$

after (children): $H_{\text{after}} = \sum_{i=1}^{k} \frac{p_i + n_i}{p + n} I\left(\frac{p_i}{p_i+n_i}, \frac{n_i}{p_i+n_i}\right)$

gain: $IG = H_{\text{before}} - H_{\text{after}}$

Pick the feature with the **largest** information gain at every node.

Jeeves: gain at the root (Outlook)

Training set has 9 Yes, 5 No $(p = 9,\ n = 5)$.

Q3. $H_{\text{before}} = I(9/14, 5/14) = ?$

$\Rightarrow$ **0.940** bits.

Q4. Split on **Outlook**: Sunny (2+/3−), Overcast (4+/0−), Rain (3+/2−).

$$H_{\text{after}} = \tfrac{5}{14}(0.971) + \tfrac{4}{14}(0) + \tfrac{5}{14}(0.971) = 0.694.$$

$$\Rightarrow IG(\text{Outlook}) = 0.940 - 0.694 = \mathbf{0.247}.$$

Jeeves: gain at the root (Humidity) and winner

Same $H_{\text{before}} = 0.940$. Now try splitting on Humidity instead.

Q5. Split on **Humidity**: High (3+/4−), Normal (6+/1−).

$$H_{\text{after}} = \tfrac{7}{14}(0.985) + \tfrac{7}{14}(0.591) = 0.788.$$

$$\Rightarrow IG(\text{Humidity}) = 0.940 - 0.788 = \mathbf{0.151}.$$

Q6. Which feature do we pick as the root?

Outlook — higher IG: $0.247 > 0.151$.

ID3: the canonical algorithm

`ID3(examples, features)`

1. Apply the three base cases from DT-Learner (same-class / no-features / no-examples).
2. For each feature f in features, compute $IG(f)$ on the current examples.
3. Choose $f^\star = \arg\max_f \ IG(f)$; make an internal node testing $f^\star$.
4. For each value v of $f^\star$: recurse on examples with $f^\star = v$ and features $\setminus \{f^\star\}$.

Greedy: pick the locally best feature at each step. Not always globally optimal — but fast and effective.

Another purity score: Gini & CART

Information gain uses entropy. A cheaper alternative: how often we'd **mislabel** an example drawn at random from the node.

Gini impurity

$$G = 1 - \sum_{i=1}^{k} P(c_i)^2$$

0 when pure; maximal when classes are balanced — same spirit as entropy, cheaper to compute (no logs).

CART

The tree learner behind scikit-learn: **binary** splits, Gini (or entropy) for classification, squared error for regression.

Entropy vs. Gini rarely changes the tree much — both reward pure children.

Trees overfit — so we prune

Grown to purity, a tree memorizes the training set (one leaf per noisy example) and generalizes poorly — a first taste of **overfitting** (we'll formalize it next lecture).

Pre-pruning

Stop growing early: cap **max depth**, require a **minimum #examples** per node, or a minimum gain to split.

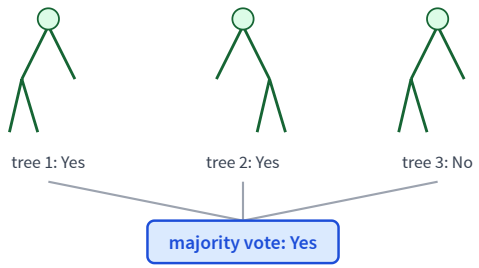
Post-pruning

Grow the full tree, then **collapse** branches that don't improve validation accuracy.

Pick the depth / pruning strength with **cross-validation** (formalized in L17).

Random forests: a crowd of trees

One deep tree is high-variance. **Bagging** averages many trees, each trained on a bootstrap sample; a **random forest** also splits on a random feature subset to decorrelate them.



- **Classification:** majority vote.
- **Regression:** average the predictions.
- Averaging cancels individual trees' errors → **lower variance**, strong off-the-shelf accuracy.

Gradient boosting: fix your own mistakes

Instead of averaging independent trees, **add trees in sequence**, each one correcting the residual errors of the trees so far:

$$F_m(x) = F_{m-1}(x) + \gamma\, h_m(x)$$

- h_m is a small tree fit to the **gradient of the loss** (for squared loss, the residuals).
- γ is a small learning rate — many shallow trees, added slowly.
- **XGBoost / LightGBM** are the go-to winners on **tabular** data today.

Forests reduce *variance* (parallel); boosting reduces *bias* (sequential).

Learning goals (recap) — Next: supervised learning

✓ Describe the **components** of a decision tree.

✓ **Construct** a tree given a feature-testing order.

✓ Compute **test-set accuracy**.

✓ Compute **entropy** and **information gain**; trace the **DT learner**.

✓ Use **Gini** / **CART**; control overfitting with **pruning**.

✓ Combine trees: **random forests** & **gradient boosting**.

L17: the general recipe behind supervised learning — **regression, loss functions, and generalization**.